

Ch. Pavan Kumar Reddy

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[GitHub](#) | [LeetCode](#) | [Codeforces](#) | [LinkedIn](#)

Education

Sardar Vallabhbhai National Institute of Technology, Surat

Bachelor of Technology in Artificial Intelligence

Aug 2024 – May 2028 (Expected)

CGPA: 8.57 (Till Sem 4)

Projects

CyberPitch: FIFA Phishing Detection Platform | [GitHub](#)

Jul 2026

Technologies: Next.js, React, TypeScript, Python, FastAPI, scikit-learn, SQLite, Playwright

- Built a **full-stack AI cybersecurity platform** detecting phishing domains impersonating FIFA/World Cup 2026 brands using a **dual-stage ML pipeline** (Random Forest + HistGradientBoosting) with SHAP-based risk explanations.
- Engineered a **concurrent scan pipeline** with **FastAPI + asyncio** orchestrating 6+ parallel OSINT services — WHOIS, DNS, SSL, HTML analysis, URLHaus blacklist, and ASN lookup — backed by a **SQLite caching layer**.
- Implemented **autonomous domain discovery** via a background scheduler querying **Certificate Transparency logs (crt.sh)** and generating typosquatting variants to proactively surface suspicious domains.
- Designed a **visual similarity engine** using **Playwright** headless screenshots and **perceptual hashing (pHash)** to detect FIFA brand impersonation at the pixel level.

Non-Invasive Anemia Detection Using Deep Learning | [GitHub](#)

2026

Technologies: Python, TensorFlow, Keras, OpenCV, EfficientNetBo, Scikit-Learn

- Built a **multi-modality deep learning pipeline** using **EfficientNetBo** with a **two-phase transfer learning strategy** to detect anemia non-invasively from **fingernail and retinal images**, eliminating the need for blood tests.
- Applied **CLAHE** (Contrast Limited Adaptive Histogram Equalization) on the **LAB color space** to enhance nail-bed color contrast, enabling the model to distinguish hemoglobin-deficient samples with greater precision.
- Handled **class imbalance** via computed class weights and implemented **data augmentation**, alongside **EarlyStopping**, **ReduceLROnPlateau**, and **label smoothing** for stable convergence.

Urban Traffic Demand Forecasting System | [GitHub](#)

june 2026

Technologies: Python, LightGBM, XGBoost, CatBoost, Optuna, PyGeohash

- Built a **traffic demand forecasting system** using an optimized ensemble of **LightGBM, XGBoost, and CatBoost** models, achieving **91.93% accuracy** and selection for **Round 2 among 10,600+ teams** in the **Flipkart Gridlock Hackathon**.
- Engineered complex **spatial-temporal features**, leveraging **PyGeohash** for coordinate extraction, **trigonometric time embeddings**, and **K-Means clustering** to capture localized geographic traffic patterns.
- Implemented an **end-to-end machine learning pipeline**, orchestrating robust data preprocessing (**KNNImputer**), strict **time-series validation** to prevent data leakage, and hyperparameter tuning with **Optuna**.

Technical Skills

Languages: Python, C, C++, MySQL, HTML, CSS, JavaScript

Web Technologies: React JS, Node.js, Express.js, MongoDB (MERN Stack)

AI/ML: Scikit-Learn, NumPy, Pandas, Machine Learning

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Automata Theory

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code

Achievements

- Runner-Up — Code Cup 2K26 (Cybersecurity Track)** | Secured **2nd place** among competing teams in a hackathon conducted by **ACM**, by building **CyberPitch**, an AI-powered phishing detection platform targeting FIFA/World Cup 2026 brand impersonation.
- LeetCode** | Solved **315+ algorithmic problems** and actively competed in 10+ global contests, achieving a peak contest rating of **1637**.
- Codeforces** | Solved **200+ competitive programming challenges** and achieved a peak rating of **1223** (Pupil rank) by optimizing data structures and algorithms under time constraints.